UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 009 · Particular integral when X is any other form · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·5	Unit 2 deck 009 (“Particular integral when X is any other form”, SESSION 7), slide 5 (rendered .pptx)
T009 00:03:13	Unit 2 AV-summary transcript “Particular Integral when X is any other form” (15.2 min), segment start
T009b 00:01:10	Unit 2 AV-summary transcript “Particular integral when X is any other form_Continued” (10.2 min), segment start
CW U2 P28	Unit 2 class-work problem 28 (pointer only; solved in the class 011 VoP pack)
ESA Dec-23 3c	Past paper: UE22MA141A Dec 2023, question 3.c

Contents

1. Figure G1.1 · $\frac{1}{D^2}\ln x$: integrate twice
2. Figure G2.1 · $(D^2+1)y=\tan x$: the PI stays bounded where $\tan x$ blows up
3. Figure G3.1 · ESA Dec-23 3c: $y_p = e^{-2x}e^{e^x}$ and general solutions
4. Figure G3.2 · $(D+1)(D+2)y=\sin(e^x)$: a chirp in, a damped chirp out

The deck has no graphs, so every figure is added for clarity. Colours follow the Okabe–Ito order: blue = the particular integral, vermilion (dashed) = the right-hand side X, green = an intermediate integral; grey for guides, asymptotes and sample general solutions. Every plotted PI was checked by substituting it back with sympy.

1 The general method: linear factors, partial fractions and $\frac{1}{D-m}X$

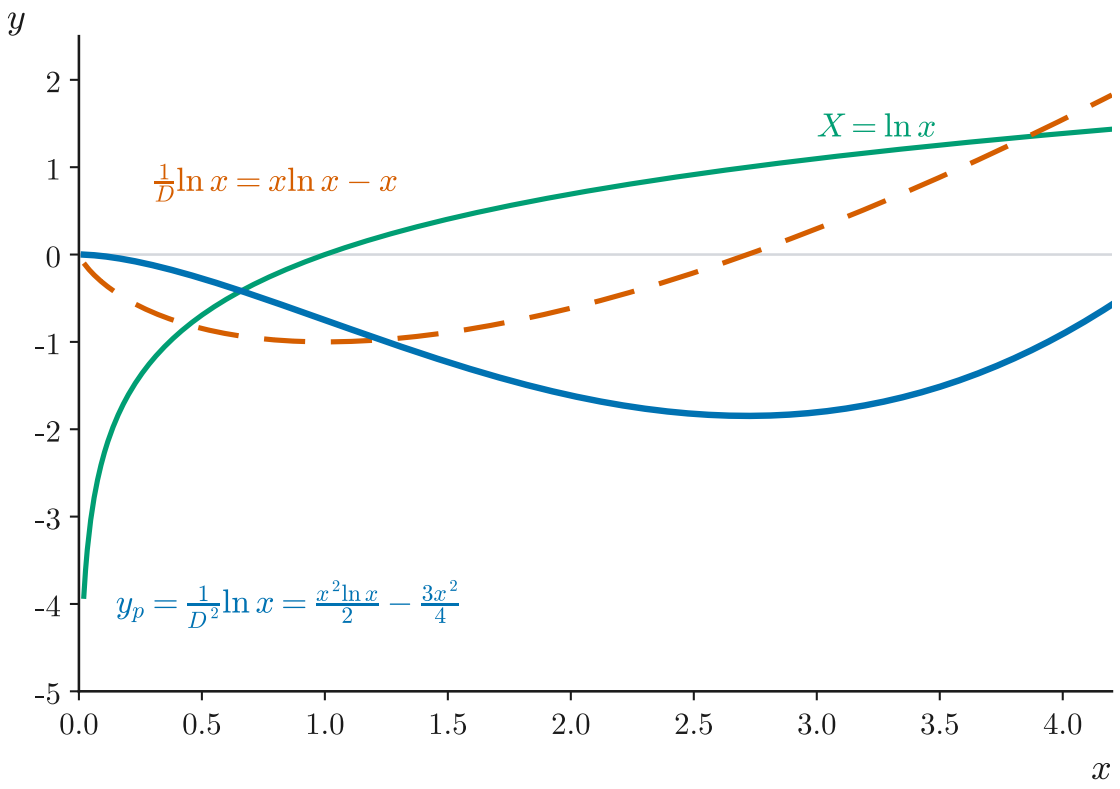


Figure G1.1. $\frac{1}{D^2}\ln x$: integrate twice. ESA ★

ADDED FOR CLARITY S009·8 · S009·9

How to read it. Green: the right-hand side $\ln x$. Vermilion dashed: one integration, $\frac{1}{D}\ln x = x\ln x - x$ (its slope is $\ln x$). Blue: the second integration, the PI $\frac{x^2\ln x}{2} - \frac{3x^2}{4}$, whose slope is the vermilion curve and whose curvature is the green one. Each curve has a turning point where the one below it crosses zero ($x = 1$ for $x\ln x - x$, $x = e$ for the PI).

2 Complex linear factors: $(D^2 + a^2)y = \tan ax$

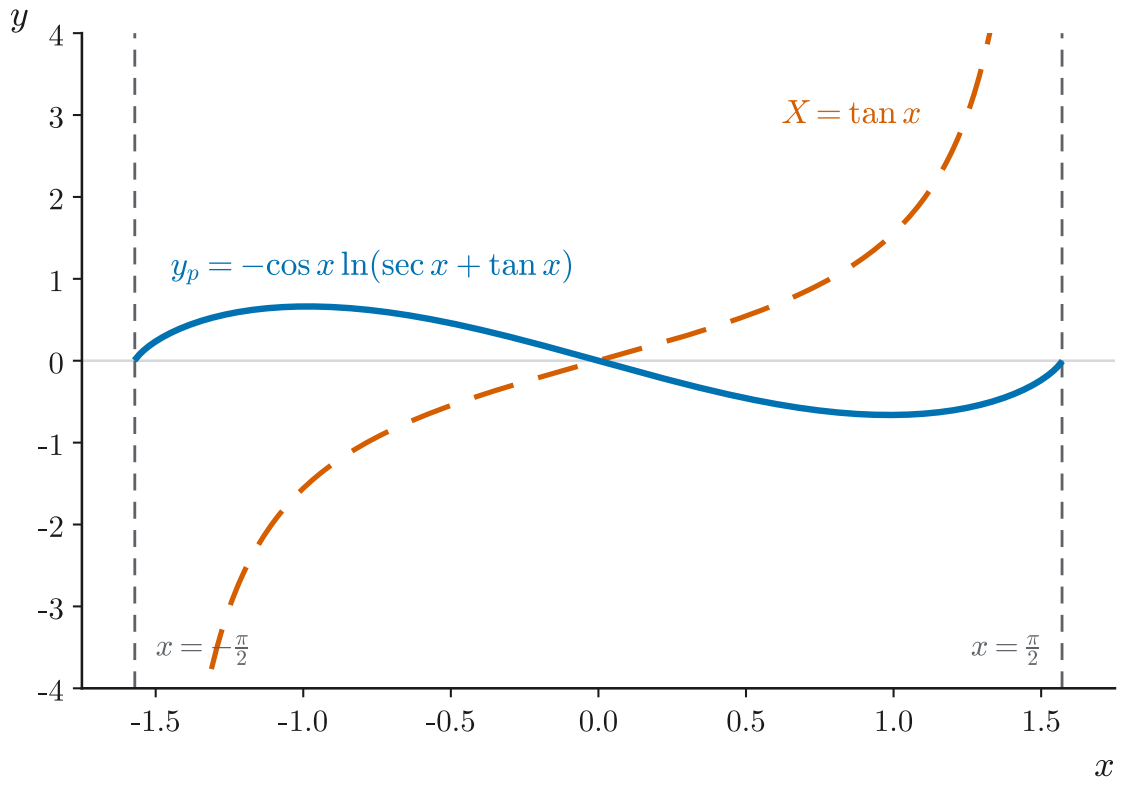


Figure G2.1. $(D^2 + 1)y = \tan x$: the PI stays bounded where $\tan x$ blows up. ESA ★★

ADDED FOR CLARITY S009·5 · S009·7

How to read it. Slide 009·7 with $a = 1$. Vermilion dashed: $X = \tan x$, with asymptotes at $x = \pm\frac{\pi}{2}$ (grey). Blue: the PI $-\cos x \ln(\sec x + \tan x)$; it is odd, like $\tan x$, and tends to 0 at $x = \pm\frac{\pi}{2}$ (where $\cos x \to 0$ beats the logarithm) although X is unbounded there.

3 Right-hand sides in e^x: the substitution $t = e^x$

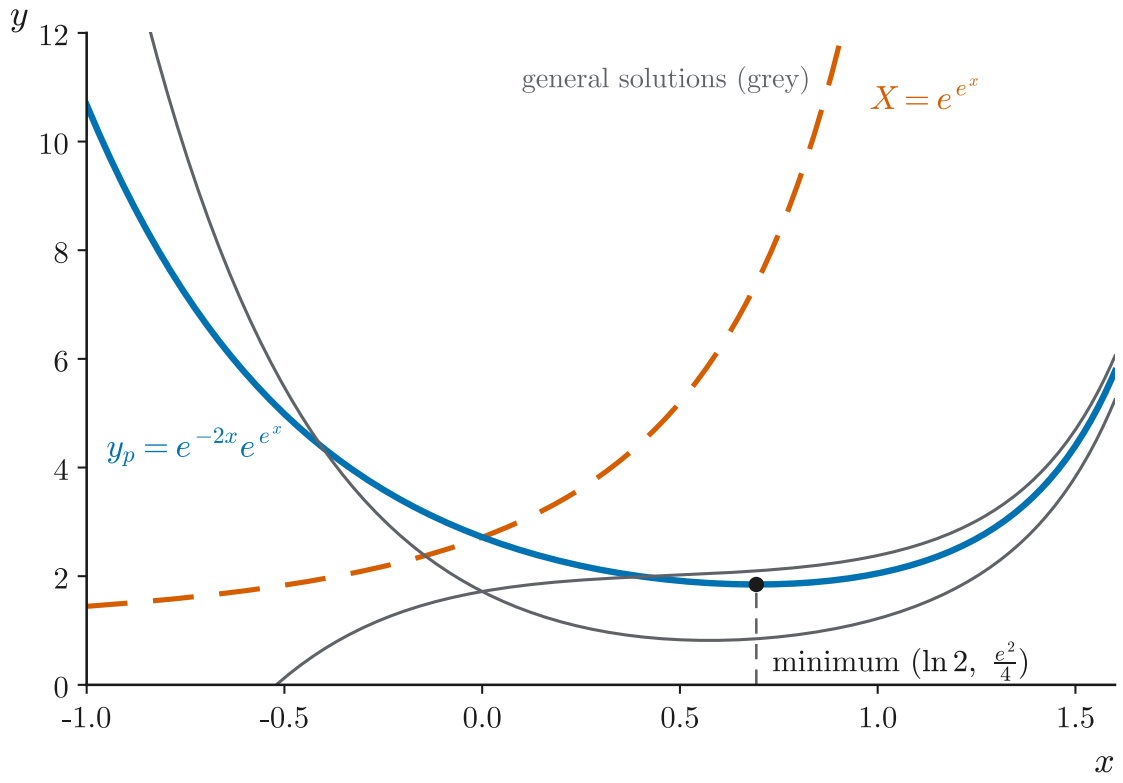


Figure G3.1. ESA Dec-23 3c: $y_p = e^{-2x}e^{e^x}$ and general solutions. ESA ★★

ADDED FOR CLARITY ESA Dec-23 3c · S009·11

How to read it. Vermilion dashed: $X = e^{e^x}$. Blue: the PI $e^{-2x}e^{e^x}$; its minimum is where $(e^x - 2x)' = e^x - 2 = 0$, at $x = \ln 2$, value $\frac{e^2}{4}$. Grey: two general solutions $c_1e^{-x} + c_2e^{-2x} + y_p$ ($c_1, c_2 = 2, -3$ and $-3, 2$); the CF dies out for large x, so every solution approaches y_p there; note $y_p = e^{-2x}X$ exactly.

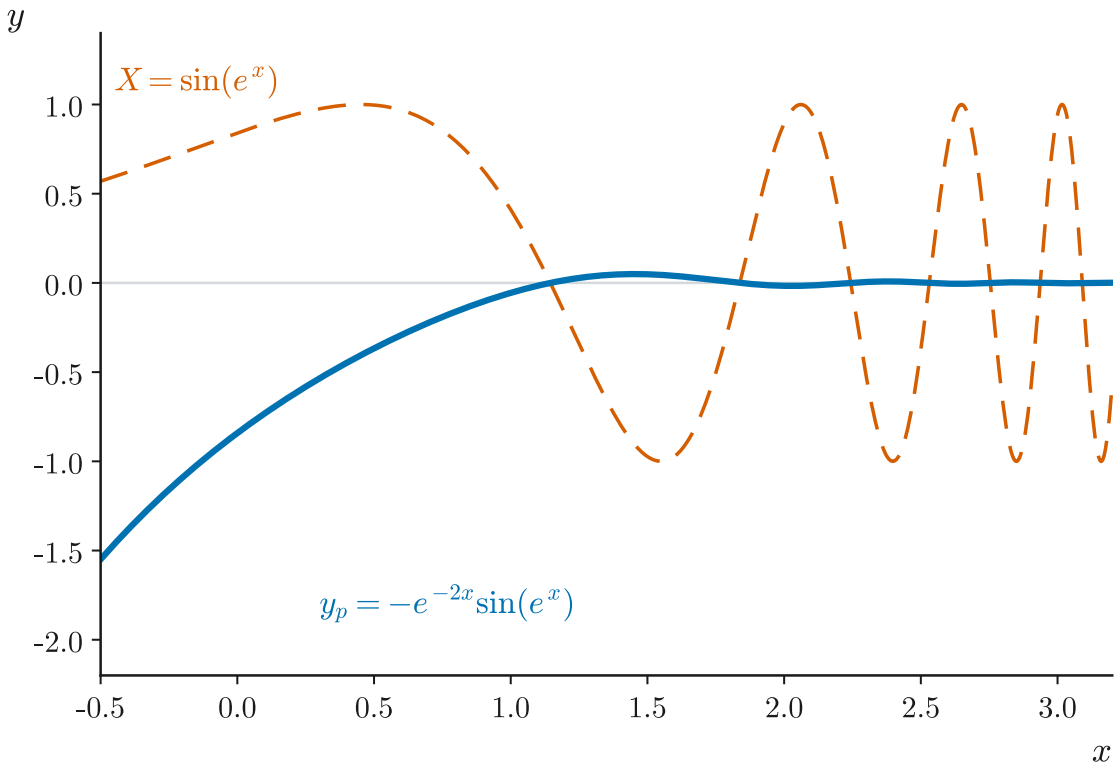


Figure G3.2. $(D+1)(D+2)y = \sin(e^x)$: a chirp in, a damped chirp out. ESA ★★

ADDED FOR CLARITY S009·12 · S009·13

How to read it. Vermilion dashed: $X = \sin(e^x)$, which oscillates faster and faster as x grows. Blue: the PI $-e^{-2x}\sin(e^x)$: the same zeros ($e^x = k\pi$) but an amplitude e^{-2x} that dies away, because $f(D)$ amplifies fast oscillations (D^2 dominates), so their PI is small.